Atomic Capacity and the Entropy Rate of Causal Simulation

A sharp feasible-region theorem for bounded output alphabets

Jan Mikulík

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Abstract

We relate the atomic capacity of an infinite controlled tree to the growth of its optimal finite-horizon causal seed entropy. If every response has at most $d \geq 2$ values, atomic capacity a and normalized upper entropy rate R satisfy $0 \leq R \leq 1 - a$. Every pair in this triangle occurs, already for a process with one action and with an actual limiting rate. The proof yields a finite-horizon bound in terms of the limiting information envelope. In particular, existence of a countably supported exact common seed forces zero entropy rate, even if every such seed has infinite entropy. The converse fails. These are deductions from the finite causal Shannon bound and the atomic-capacity identity; historical priority and Lean formalization of the deductions are not asserted.

1 Setting and imported results

Consider a locally finite infinite controlled tree. At each history, a policy chooses an available action and receives an output from the prescribed conditional law. Policies are deterministic and adaptive. Every local output alphabet has at most d elements, where $d \geq 2$ is a fixed integer. There is no uniform bound on the number of available actions. We allow zero transition probabilities. Equivalently, one may delete zero-probability outputs when computing transcript laws and extend strategies arbitrarily at null histories.

A common causal seed is a random variable whose value fixes one total deterministic response strategy, shared by all policies. For the depth-n tree, let C_n be the infimum of seed Shannon entropy among exact common causal realizations. All entropies and information variables below are in bits. Finite response tables suffice at finite horizons.

For policy π, let $P_{n,\pi}$ be its transcript law and put

$$I_{n,\pi} = -\log_2 P_{n,\pi}(Y), \qquad F_n^*(t) = \inf_\pi \Pr(I_{n,\pi} \leq t), \qquad L_n = \int_0^\infty (1 - F_n^*(t))\, dt.$$

The information variable is evaluated only on positive-mass transcripts. The finite infimum over horizon-n policies is a minimum. Define

$$F_\infty(t) = \inf_{n \geq 1} F_n^*(t), \qquad a = \mathrm{atomQ} = \sup_{K \geq 1} \inf_{n \geq 1, \pi} A_K(P_{n,\pi}),$$

where $A_K(P)$ is the mass of the K largest atoms, with all atoms used if there are fewer than K.

We use the following results of the causal Shannon framework [1]:

$$L_n \leq C_n \leq L_n + c_\star, \qquad c_\star = \frac{1 + \log_2 e}{2}, \tag{1}$$

$$\lim_{t \to \infty} F_\infty(t) = a, \tag{2}$$

$$\text{an exact countable common seed exists} \iff a = 1. \tag{3}$$

The capacity theorem also states that the atomic mass of every exact common source is at most a, and that a causal strategy law attains this capacity. These statements are applied in their zero-branch-compatible form. No finite-entropy assumption on an infinite source is imposed.

2 A finite-horizon capacity bound

Theorem 1. *For every integer $n \geq 1$,*

$$\boxed{C_n \leq \int_0^{n \log_2 d} (1 - F_\infty(t))\, dt + \log_2 e + c_\star.} \tag{4}$$

Consequently, with $R = \limsup_{n\to\infty} C_n/(n \log_2 d)$,

$$\boxed{0 \leq R \leq 1 - a.} \tag{5}$$

Proof. Fix a deterministic policy. At horizon n its output histories, and hence its action–output transcripts, number at most d^n: the actions are determined recursively by the observed history. Every transcript contributing to $\{I_{n,\pi} > t\}$ has mass less than 2^{-t}. Therefore

$$\Pr(I_{n,\pi} > t) \leq d^n 2^{-t}.$$

Taking the supremum over policies proves

$$1 - F_n^*(t) \leq \min\{1, d^n 2^{-t}\}.$$

Also $F_n^*(t) \geq F_\infty(t)$ by definition. With $b_n = n \log_2 d$,

$$\begin{aligned} L_n &= \int_0^{b_n} (1 - F_n^*(t))\, dt + \int_{b_n}^{\infty} (1 - F_n^*(t))\, dt \\ &\leq \int_0^{b_n} (1 - F_\infty(t))\, dt + \int_{b_n}^{\infty} d^n 2^{-t}\, dt \\ &= \int_0^{b_n} (1 - F_\infty(t))\, dt + \log_2 e. \end{aligned}$$

Applying (1) gives (4). The bounded measurable function $1 - F_\infty$ converges to $1 - a$ by (2). Hence its Cesàro averages satisfy

$$\frac{1}{b_n} \int_0^{b_n} (1 - F_\infty(t))\, dt \longrightarrow 1 - a.$$

Divide (4) by $b_n \to \infty$ and take the upper limit. Nonnegativity of entropy supplies the lower bound. □

3 Every point of the triangle occurs

Theorem 2. *For each integer $d \geq 2$ and each pair (a, r) satisfying $0 \leq a \leq 1$ and $0 \leq r \leq 1-a$, there is a one-action process with at most d outputs per time, atomic capacity a, and*

$$\lim_{n\to\infty} \frac{C_n}{n \log_2 d} = r.$$

Thus (5) *describes the entire feasible region, also when the rate is required to exist as a limit.*

Proof. The case $a = 1$ necessarily has $r = 0$ and is realized by a deterministic process. Suppose $a < 1$ and set $\theta = r/(1-a) \in [0,1]$. Choose an infinite deterministic set $D \subseteq \{2,3,\ldots\}$ whose asymptotic density is θ. For $\theta > 0$, take the shifted indices $k+1$ for which $\lfloor \theta k \rfloor - \lfloor \theta(k-1) \rfloor = 1$. For $\theta = 0$, take $D = \{2^k : k \geq 1\}$.

The first output is a branch indicator: it equals 0 with probability a and 1 with probability $1-a$. On branch 0, all later outputs are 0. On branch 1, the outputs at times in D are independent and uniform on $\{0,\ldots,d-1\}$; all other outputs are 0. This is a controlled tree with only one action, with deterministic transitions allowed.

Write $m_n = |D \cap \{2,\ldots,n\}|$. The length-$n$ law consists of one atom of mass a and d^{m_n} atoms of mass $(1-a)d^{-m_n}$, with any zero-mass atom omitted. Thus

$$C_n = H(P_n) = h_2(a) + (1-a)m_n \log_2 d. \tag{6}$$

Indeed, every exact seed maps deterministically to the transcript, giving the entropy lower bound. Conversely, the transcript itself is a seed: its successive symbols define the response along its path, with arbitrary responses on histories off that path. Hence the lower bound is attained.

For completeness, the capacity can be checked directly without an identification theorem. For each $K \geq 1$, $A_K(P_n) \geq a$ for every n, whereas

$$A_K(P_n) \leq a + K(1-a)d^{-m_n}.$$

As D is infinite, $m_n \to \infty$. It follows that $\inf_n A_K(P_n) = a$, and then $\sup_K \inf_n A_K(P_n) = a$. Finally $m_n/n \to \theta$, so (6) gives the claimed rate. □

4 Consequences and limitations

Corollary 3 (Countable realization forces zero rate)**.** *If one countably supported causal seed is exact at every finite horizon, then $C_n = o(n)$. This remains true when all such infinite-horizon seeds have infinite Shannon entropy.*

Proof. By (3), $a = 1$. Theorem 1 gives $R = 0$; nonnegativity then implies $C_n/(n \log_2 d) \to 0$. □

Corollary 4 (Zero rate does not imply countable realization)**.** *There is a binary one-action process with $C_n = \log_2 n + O(1)$ and no countably supported exact infinite-horizon seed.*

Proof. Generate independent fair bits at times 2^k, $k \geq 1$, and output zero otherwise. The entropy through time $n \geq 2$ is exactly $\lfloor \log_2 n \rfloor$. Each infinite transcript has probability zero, since it specifies infinitely many fair bits. A deterministic image of a countably supported seed cannot have this non-atomic law. □

Corollary 5 (A lower bound on the required non-atomic mass)**.** *If the normalized upper cost rate is R, every exact common source has atomic mass at most $1-R$. For a probability measure on the strategy space, its non-atomic component therefore has mass at least R.*

Proof. The capacity theorem bounds the atomic mass by a, and Theorem 1 yields $a \leq 1-R$. □

The finite-horizon seeds attaining or approximating C_n may depend on n. The zero-rate conclusion does not construct one finite-entropy seed for the whole infinite tree. It also does not bound computational time or the description length of the kernels and decoders.

The sharpness examples use zero transition probabilities. No claim is made here that every boundary point is realized under a requirement that every binary transition probability be strictly between zero and one. The constant in (4) is a valid uniform bound, not a claim of optimal additive overhead.

5 Relation to classical theory and verification status

The relation between atomic components and normalized entropy growth has classical antecedents in information dimension; see, for example, Wu [2]. The specific deduction here concerns the minimum entropy of one causal seed for all adaptive policies and uses their common lower information envelope. The universal estimate (4) and the sharp feasible-region formulation are the statements to compare with prior work. We do not assert a new discovery of the underlying discrete/non-atomic distinction.

The proofs above depend on (1)–(3) from the causal Shannon framework. They are ordinary mathematical proofs, not new Lean-kernel certificates. Formalizing these corollaries requires connecting the bounded transcript count, the tail integral, and the capacity limit to the repository's exact definitions.

References

[1] J. Mikulík, *One Seed for Every Adaptive Policy*, Lean development and mathematical notes. `https://github.com/jendamikulik/finite-Shannon-theorem`.

[2] Y. Wu, *Shannon Theory for Compressed Sensing*, Ph.D. dissertation, Princeton University, 2011. `https://www.stat.yale.edu/~yw562/reprints/wu.thesis.pdf`.